Strict Univalent Universes of an $(\infty, 1)$-Topos are Essentially Unique

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What Does a Topos Model?

1-Topoi

A (Grothendieck) 1-topos $\mathcal{E}$ has an internal logic: it models a form of intuitionistic **set theory**. Objects behave like sets, subobjects like predicates, the subobject classifier Ω like truth values.

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An ∞-topos $\mathcal{E}_\infty$ is the higher-categorical analogue. What does *it* model, via its own internal logic?

Answer: a form of **Homotopy Type Theory** – types as spaces, equality as paths.

Strictification

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For an internal *logic* (a type theory) to make sense, we need a setting with genuine, on-the-nose equalities.

Strategy: present $\mathcal{E}_\infty$ by an honest 1-category – a model category $\mathcal{E}$ – where composition really is associative, and study the type theory *there*.

Roadmap

- **Univalent universes of ∞-topoi are essentially unique:**

1. Background
2. Univalent Universes
3. Shulman's Construction
4. Uniqueness

Strict Presentations of ∞-Topoi

Theorem 4.18 (Gepner–Kock)

Every presentable, locally cartesian closed ∞-category C – in particular every ∞-topos – is presented by a combinatorial type-theoretic model category M:

$$C \simeq N(M)[W^{-1}],$$

and M can be taken **right proper**.

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From now on we work inside such an $\mathcal{E}$ – this is exactly the strictification promised above.

Cisinski Model Categories

A type theoretic model topos is in particular a right proper Cisinski's model category.

Definition

A **Cisinski model category** is a Grothendieck topos $\mathcal{E}$ equipped with a combinatorial model structure whose cofibrations are exactly the monomorphisms.

Modeling HoTT

Fact

Homotopy Type Theory can be modeled in any **right proper** Cisinski model category $\mathcal{E}$: types as objects, terms as (generalized) elements, equality as path objects.

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The hard part

Univalence. Constructing an univalent universe for a right proper Cisinski's model category was not trivial before Shulman's theorem. Also the universe constructed is a priori not *canonical*. This axiom is the core of this talk.

The Starting Point

Shulman's Theorem (2019)

Every $(\infty, 1)$-topos $\mathcal{E}_\infty$ can be *presented* by a category with a strict univalent universe.

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Many constructions of such universes now exist.

This talk

Fix $\mathcal{E}_\infty$ and a large enough cardinal λ. *Any* two strict univalent universes for λ are canonically equivalent: they represent the **same functor**

$$X \longmapsto (\mathcal{E}_\infty^\lambda / X)^{\simeq}.$$

Interval, Path, and Cylinder Objects

Interval object (Def. 2.3)

An **interval object** is a factorization of the fold map $* \sqcup * \to *$ into a cofibration followed by a trivial fibration. In a Cisinski's model category the subobject classifier Ω is always an interval object (Ex. 2.5).

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Path object (Def. 2.9)

A **path object** for X is a factorization of the diagonal

$$X \xrightarrow{\sim} X^{\Delta^1} \twoheadrightarrow X \times X$$

into a trivial cofibration followed by a fibration.

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Cylinder object (Def. 2.15)

Dually, a **cylinder object** for X is a factorization

$$X \sqcup X \rightarrowtail I(X) \xrightarrow{\sim} X$$

into a cofibration followed by a trivial fibration.

Homotopy Between Morphisms

Let $f, g : X \to Y$.

Right homotopy (Def. 2.11)

Let $(f, g) : X \to Y \times Y$ be the map induced by the universal property of the product. Then $f \sim_R g$ if (f, g) factors as

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Left homotopy (Def. 2.17)

Let $(f, g) : X \sqcup X \to Y$ be the map induced by the universal property of the coproduct. Then $f \sim_L g$ if (f, g) factors as

$$X \sqcup X \rightarrowtail I(X) \to Y$$

through the inclusion of *some* cylinder object $I(X)$.

Homotopy is an Equivalence Relation

Proposition 2.20 / Lemma 2.22

$\sim_L$ and $\sim_R$ are **equivalence relations** on $\mathrm{Hom}(X, Y)$ and are independent on the choice of path object Y^{Δ^1} or cylinder object $I(X)$. If X is cofibrant and Y is fibrant, $\sim_L$ and $\sim_R$ agree.

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This defines the homotopy category $[X, Y] := \mathrm{Hom}(X, Y)/\sim$, and (Lemma 2.30)

$$[X, Y] \cong \mathrm{Hom}_{\mathrm{ho}(\mathcal{E})}(X, Y) \qquad (Y \text{ fibrant}).$$

Since every object of a Cisinski's model category is cofibrant, this applies to all X.

Equivalences, Internally

A Cisinski model category $\mathcal{E}$ is a topos, hence cartesian closed: it has an internal Hom $\mathcal{E}(-,-)$. For $A, B \in \mathcal{E}$ there is an object $\mathrm{Equiv}(A,B) \in \mathcal{E}$, characterized as follows.

Lemma 2.53

A map $e : X \to \mathrm{Equiv}(A,B)$ corresponds precisely to the data of

1. a map $f : X \times A \to X \times B$ over X,
2. two maps $g, h : X \times B \to X \times A$ over X,
3. homotopies $H : g \circ f \sim \mathrm{id}_{X \times A}$ and $K : f \circ h \sim \mathrm{id}_{X \times B}$.

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So $\mathrm{Equiv}(A,B)$ is exactly the object parametrizing equivalences $A \to B$ together with a two-sided homotopy inverse.

For $z : Z \to *$

$$\mathrm{Equiv}_Z(A \times Z, A \times Z) \cong z^*\mathrm{Equiv}(A,B)$$

Univalence Axiom

For a fibration $\pi : U_\bullet \twoheadrightarrow U$, define

$$\mathrm{Eq}(U_\bullet) := \mathrm{Equiv}_{U\times U}\big(U_\bullet \times U,\ U \times U_\bullet\big) \ \in\ \mathcal{E}/(U \times U),$$

Definition

π is **univalent** if $\mathrm{Eq}(U_\bullet) \to U \times U$ is (via the diagonal) a *path object* for U.

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Strict univalent universe

A univalent $\pi : U_\bullet \twoheadrightarrow U$ such that *every* λ-small fibration $A \twoheadrightarrow B$ arises as a **strict pullback** of π.

So U genuinely classifies small fibrations (not as a bijection of sets).

Univalence axiom

There exists a cardinal k such that for any inaccessible cardinal $\lambda > k$ there exists a strict univalent universe $U_\bullet \twoheadrightarrow U$ classifying λ-small fibrations.

Building the Universe: Fibered Structures

- Work in $\mathrm{Psh}(\mathcal{E}) = \mathrm{PsFun}(\mathcal{E}^{op}, \mathrm{Gpd})$.
- A **notion of a fibered structure** on $\mathcal{E}$: a discrete fibration of pseudo-presheaves $F \to \mathscr{E}$ (the self-indexing) with small fibers.
- There is a notion of fibered structure whose image are λ-small fibrations.
- An **universe** for F is a cofibrant U together with an acyclic fibration

$$\mathcal{E}(-, U) \longrightarrow F.$$

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Shulman's Theorem 5.9

If F is locally representable and a stack of cell complexes, a 2-categorical **small object argument** produces the acyclic fibration $\mathcal{E}(-, U) \to F$ needed.

Shulman generalizes this result to the image of certain notions of fibered structures.

From a Universe to Univalence

Getting existence is one thing; **univalence** needs more care. Shulman's recipe (Discussion 3.44):

1. Build $U_\bullet \to U$ classifying λ-small fibrations.
2. Show U is fibrant.
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These reduce to two structural properties of $\mathcal{E}$ itself:

3′) fibrations transport along trivial cofibrations.

4′) weak equivalences of λ-small fibrations transport along cofibrations (Theorem 3.47) – proved via right properness.

Both hold whenever $\mathcal{E}$ is right proper.

The Classifying Formula

Proposition 4.27

For $X \in \mathcal{E}$ and a strict univalent universe U_λ,

$$[X, U_\lambda] \;\cong\; \pi_0\, \pi_\infty\, \mathscr{W}\big(\mathrm{Fib}_{/X}\big)^\lambda,$$

where $\mathscr{W}\big(\mathrm{Fib}_{/X}\big)^\lambda$ is the nerve of λ-small fibrations over X with weak equivalences between them.

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Combined with a mapping-space recognition principle (Cisinski 7.10.6) and $\pi_\infty \mathscr{W}(\mathrm{Fib}_{/X})^\lambda \simeq (\mathcal{E}^\lambda_\infty/X)^{\simeq}$, this upgrades to an equivalence of *spaces*, functorially in X:

$$\mathcal{E}_\infty(X, U_\lambda) \;\simeq\; (\mathcal{E}^\lambda_\infty/X)^{\simeq}.$$

Main Theorem

Theorem 4.33 (Essential Uniqueness)

Let $\mathcal{E}_\infty$ be an ∞-topos and λ a cardinal. Any two strict univalent universes U_λ, U'_λ for λ-small fibrations are canonically equivalent.

Moreover, the anima of strict univalent universes for λ-small fibrations in $\mathcal{E}_\infty$ is a contractible space.

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Why: both represent the *same* functor $X \mapsto (\mathcal{E}_\infty^\lambda / X)^\simeq$, so Yoneda for ∞-categories finishes the proof.

Summary

- An ∞-topos should model an internal, HoTT-style logic; making sense of **strict equalities** forces us to **strictify** – pass to a presenting model category (Gepner–Kock).
- Homotopy Type Theory is modeled in any right proper Cisinski model category; the hard part of that is **univalence**.
- Shulman: such a category always carries a strict univalent universe, built via a 2-categorical small object argument.
- The classification formula $[X, U_\lambda] \cong \pi_0\pi_\infty \mathscr{W}(\mathrm{Fib}_{/X})^\lambda$ upgrades to a space-level equivalence.
- Consequently: **strict univalent universes are essentially unique** – the space of choices is contractible, and this is **independent on the chosen presentation**.

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Thank you!